Lab11

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```r
library(bio3d)

pdb_files <- list.files("HIV_dimer_23119", pattern = ".pdb$")

pdb <- read.pdb(paste0("HIV_dimer_23119/", pdb_files[1]))

pdb
```

```
 Call:  read.pdb(file = paste0("HIV_dimer_23119/", pdb_files[1]))

   Total Models#: 1
     Total Atoms#: 1514,  XYZs#: 4542  Chains#: 2  (values: A B)

     Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 0  (residues: 0)
     Non-protein/nucleic resid values: [ none ]

   Protein sequence:
      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
      VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, calpha, call
```

There are two proteins that are included chains A and B with a total of 198 residues and no non-protein atoms, shows AlphaFold model succesffuly predicting the structure.

Make a vector of input PDB file names that we can read into

```r
pdbfiles <- list.files("HIV_dimer_23119",
           pattern = ".pdb",
           full.names =TRUE)
pdbfiles
```

```
[1] "HIV_dimer_23119/HIV_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_0
[2] "HIV_dimer_23119/HIV_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_0
[3] "HIV_dimer_23119/HIV_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_0
[4] "HIV_dimer_23119/HIV_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_0
[5] "HIV_dimer_23119/HIV_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_0
```

```r
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```r
library(bio3d)

# Read all data from Models
#  and superpose/fit coords
pdbs <- pdbaln(pdbfiles, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
HIV_dimer_23119/HIV_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pd
HIV_dimer_23119/HIV_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pd
HIV_dimer_23119/HIV_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pd
HIV_dimer_23119/HIV_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pd
HIV_dimer_23119/HIV_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pd
.....

Extracting sequences

pdb/seq: 1   name: HIV_dimer_23119/HIV_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_
pdb/seq: 2   name: HIV_dimer_23119/HIV_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_
pdb/seq: 3   name: HIV_dimer_23119/HIV_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_
pdb/seq: 4   name: HIV_dimer_23119/HIV_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_
pdb/seq: 5   name: HIV_dimer_23119/HIV_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_
```

All five AlphaFold models show the same sequence with no gaps in the alighnment.

```r
pdbs
```

```
                               1        .         .         .         .         50
[Truncated_Name:1]HIV_dimer_   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]HIV_dimer_   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]HIV_dimer_   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]HIV_dimer_   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]HIV_dimer_   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
                               **************************************************
                               1        .         .         .         .         50

                              51        .         .         .         .         100
[Truncated_Name:1]HIV_dimer_   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]HIV_dimer_   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]HIV_dimer_   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]HIV_dimer_   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]HIV_dimer_   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
                               **************************************************
                              51        .         .         .         .         100

                             101        .         .         .         .         150
[Truncated_Name:1]HIV_dimer_   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:2]HIV_dimer_   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:3]HIV_dimer_   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:4]HIV_dimer_   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:5]HIV_dimer_   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
                               **************************************************
                             101        .         .         .         .         150

                             151        .         .         .         .       198
[Truncated_Name:1]HIV_dimer_   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]HIV_dimer_   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]HIV_dimer_   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]HIV_dimer_   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]HIV_dimer_   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
                               ************************************************
                             151        .         .         .         .       198

Call:
  pdbaln(files = pdbfiles, fit = TRUE, exefile = "msa")

Class:
```

```
  pdbs, fasta

Alignment dimensions:
  5 sequence rows; 198 position columns (198 non-gap, 0 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

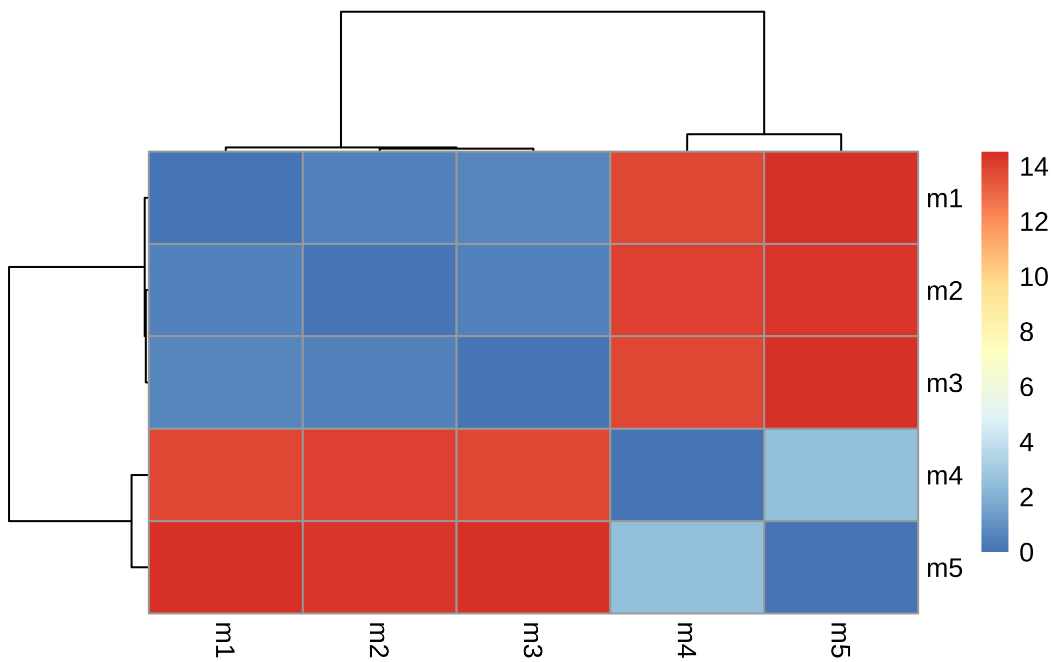
```r
rd <- rmsd(pdbs, fit=T)
```

```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

```r
range(rd)
```

```
[1]  0.000 14.526
```

```r
library(pheatmap)
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```



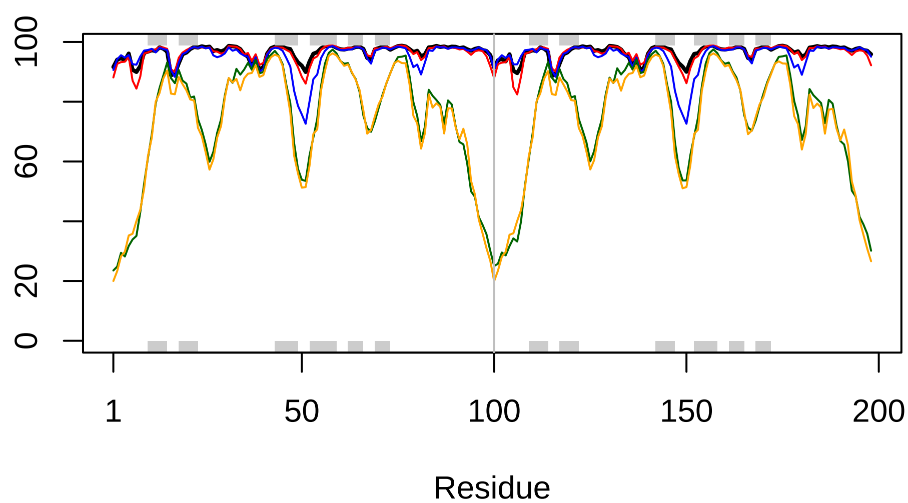
the heatmap reveals how clustered the structurally similar models.Models 1-3 seem more related to one another. They may show different conformations for the dimer. Models 4-5 are similar to each other and are more disimilar to models 1 and 2 than 3.

```
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```
Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
/var/folders/h2/hfj5mrrj37j2yphr_r6brjnh0000gn/T//RtmptxguT6/1hsg.pdb exists.
Skipping download
```

```
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```



to find the consistent "rigid core" across all models `core.find` is used.

```r
core <- core.find(pdbs)
```

```
core size 197 of 198  vol = 5437.294
core size 196 of 198  vol = 4705.336
core size 195 of 198  vol = 1827.704
core size 194 of 198  vol = 1121.539
core size 193 of 198  vol = 1047.76
core size 192 of 198  vol = 999.32
core size 191 of 198  vol = 953.718
core size 190 of 198  vol = 910.755
core size 189 of 198  vol = 870.203
core size 188 of 198  vol = 836.304
core size 187 of 198  vol = 805.237
core size 186 of 198  vol = 775.99
core size 185 of 198  vol = 752.564
core size 184 of 198  vol = 712.023
core size 183 of 198  vol = 685.568
core size 182 of 198  vol = 663.911
core size 181 of 198  vol = 645.881
core size 180 of 198  vol = 627.97
core size 179 of 198  vol = 611.812
core size 178 of 198  vol = 595.931
core size 177 of 198  vol = 581.132
core size 176 of 198  vol = 566.736
core size 175 of 198  vol = 548.587
core size 174 of 198  vol = 534.114
core size 173 of 198  vol = 505.214
core size 172 of 198  vol = 491.225
core size 171 of 198  vol = 473.905
core size 170 of 198  vol = 460.426
core size 169 of 198  vol = 444.81
core size 168 of 198  vol = 431.661
core size 167 of 198  vol = 421.542
core size 166 of 198  vol = 405.601
core size 165 of 198  vol = 392.666
core size 164 of 198  vol = 381.077
core size 163 of 198  vol = 367.559
core size 162 of 198  vol = 358.379
core size 161 of 198  vol = 346.865
core size 160 of 198  vol = 334.809
core size 159 of 198  vol = 324.09
core size 158 of 198  vol = 312.153
```

```
core size 157 of 198  vol = 301.296
core size 156 of 198  vol = 290.431
core size 155 of 198  vol = 281.319
core size 154 of 198  vol = 272.529
core size 153 of 198  vol = 263.215
core size 152 of 198  vol = 253.54
core size 151 of 198  vol = 240.86
core size 150 of 198  vol = 227.447
core size 149 of 198  vol = 215.581
core size 148 of 198  vol = 202.041
core size 147 of 198  vol = 195.426
core size 146 of 198  vol = 188.721
core size 145 of 198  vol = 181.778
core size 144 of 198  vol = 173.615
core size 143 of 198  vol = 165.946
core size 142 of 198  vol = 156.117
core size 141 of 198  vol = 149.814
core size 140 of 198  vol = 143.616
core size 139 of 198  vol = 135.81
core size 138 of 198  vol = 127.851
core size 137 of 198  vol = 122.596
core size 136 of 198  vol = 117.203
core size 135 of 198  vol = 109.848
core size 134 of 198  vol = 104.812
core size 133 of 198  vol = 98.776
core size 132 of 198  vol = 94.799
core size 131 of 198  vol = 90.494
core size 130 of 198  vol = 87.403
core size 129 of 198  vol = 83.558
core size 128 of 198  vol = 79.08
core size 127 of 198  vol = 75.056
core size 126 of 198  vol = 71.238
core size 125 of 198  vol = 67.735
core size 124 of 198  vol = 64.289
core size 123 of 198  vol = 61.381
core size 122 of 198  vol = 57.515
core size 121 of 198  vol = 53.254
core size 120 of 198  vol = 48.654
core size 119 of 198  vol = 45.832
core size 118 of 198  vol = 41.819
core size 117 of 198  vol = 38.71
core size 116 of 198  vol = 36.294
core size 115 of 198  vol = 33.386
```

```
 core size 114 of 198  vol = 30.472
 core size 113 of 198  vol = 27.786
 core size 112 of 198  vol = 25.403
 core size 111 of 198  vol = 22.827
 core size 110 of 198  vol = 21.106
 core size 109 of 198  vol = 19.327
 core size 108 of 198  vol = 17.796
 core size 107 of 198  vol = 16.235
 core size 106 of 198  vol = 14.508
 core size 105 of 198  vol = 12.969
 core size 104 of 198  vol = 11.834
 core size 103 of 198  vol = 11.185
 core size 102 of 198  vol = 10.298
 core size 101 of 198  vol = 8.898
 core size 100 of 198  vol = 7.813
 core size 99 of 198  vol = 6.074
 core size 98 of 198  vol = 5.286
 core size 97 of 198  vol = 4.43
 core size 96 of 198  vol = 3.873
 core size 95 of 198  vol = 3.321
 core size 94 of 198  vol = 2.855
 core size 93 of 198  vol = 2.293
 core size 92 of 198  vol = 1.937
 core size 91 of 198  vol = 1.631
 core size 90 of 198  vol = 1.331
 core size 89 of 198  vol = 0.957
 core size 88 of 198  vol = 0.803
 core size 87 of 198  vol = 0.647
 core size 86 of 198  vol = 0.532
 core size 85 of 198  vol = 0.444
 FINISHED: Min vol ( 0.5 ) reached
```

```r
core.inds <- print(core, vol=0.5)
```

```
# 86 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     7   7      1
2     9  49     41
3    52  95     44
```

This takes the identified core atoms and writes them to a fitted structures in a directory , these superposed coordinates are written to `corefit_structures`

```
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

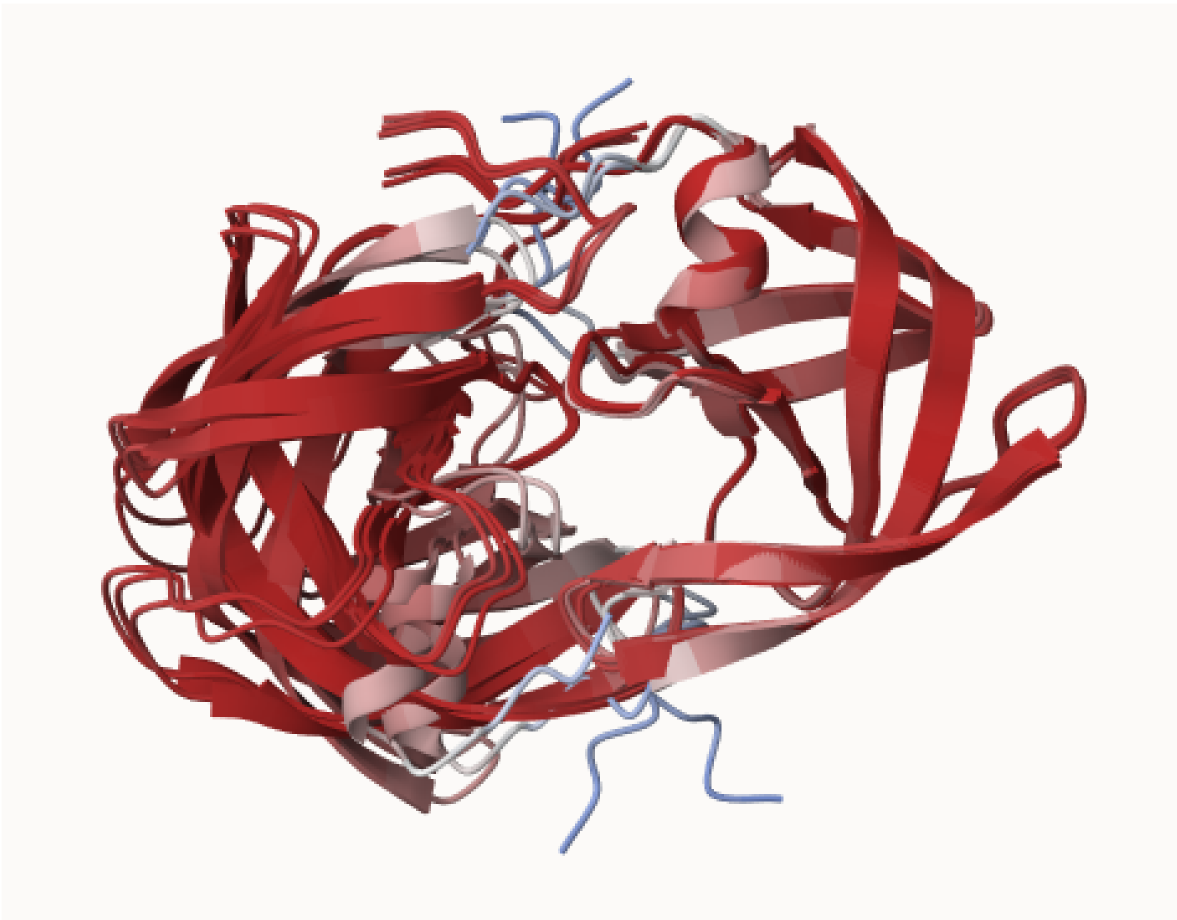


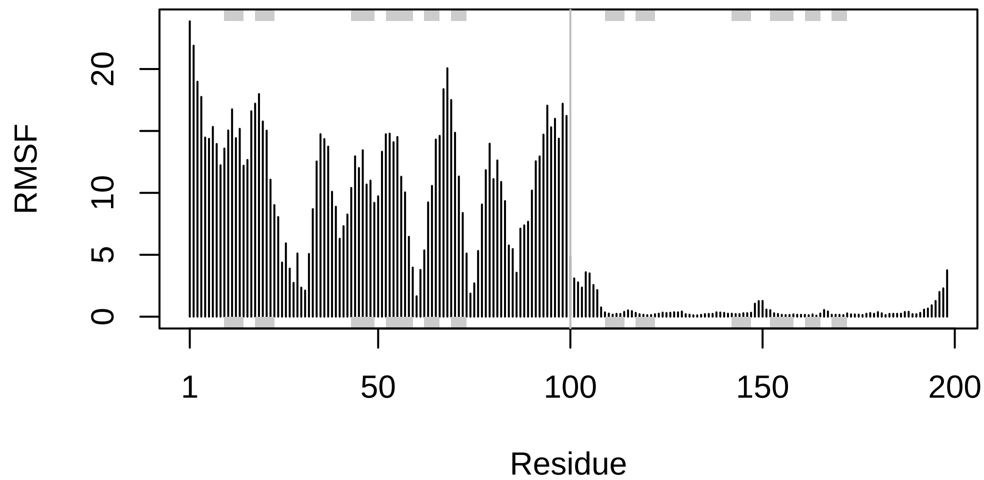
Figure 1: B factor core superposed structures

rmsf is used to measure the conformational variance, lower values demonstrate structural similarity.

```
rf <- rmsf(xyz)

#plotting rf

plotb3(rf, sse=pdb, ylab = "RMSF")
abline(v=100, col="gray")
```

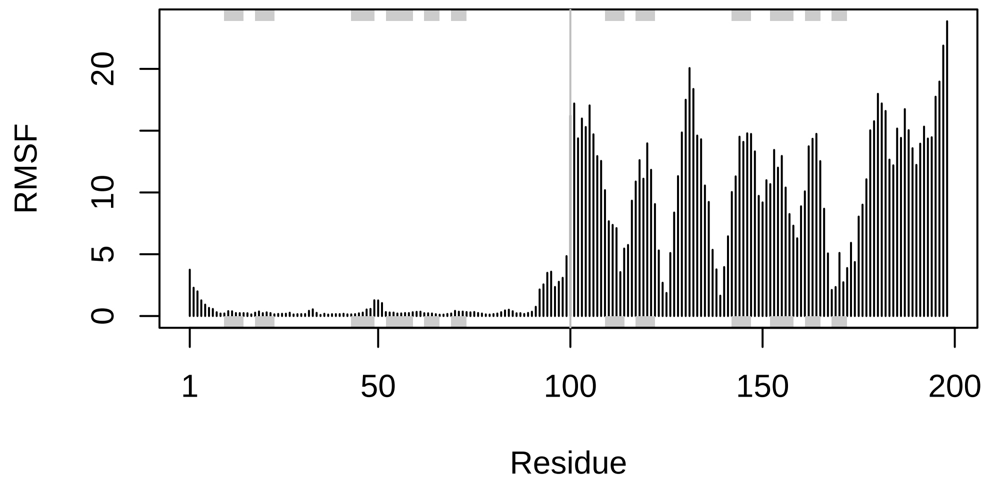


The RMSF profile is inverted compared to the lab manuel due to the ordering of the dimer chains differing in my AlphaFold predicaiton. Same trend is observed overall as one chain is rigid and the other has more variability in its conformation.

A revered RMSF vector:

```
rf <- rev(rmsf(xyz))

plotb3(rf, sse=pdb, ylab="RMSF")
abline(v=100, col="gray")
```




```r
library(jsonlite)

results_dir <- "HIV_dimer_23119/"

pae_files <- list.files(
  path = results_dir,
  pattern = ".*model.*\\.json",
  full.names = TRUE
)

pae_files
```

```
[1] "HIV_dimer_23119//HIV_dimer_23119_scores_rank_001_alphafold2_multimer_v3_model_4_seed_000
[2] "HIV_dimer_23119//HIV_dimer_23119_scores_rank_002_alphafold2_multimer_v3_model_1_seed_000
[3] "HIV_dimer_23119//HIV_dimer_23119_scores_rank_003_alphafold2_multimer_v3_model_5_seed_000
[4] "HIV_dimer_23119//HIV_dimer_23119_scores_rank_004_alphafold2_multimer_v3_model_2_seed_000
[5] "HIV_dimer_23119//HIV_dimer_23119_scores_rank_005_alphafold2_multimer_v3_model_3_seed_000
```

```r
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

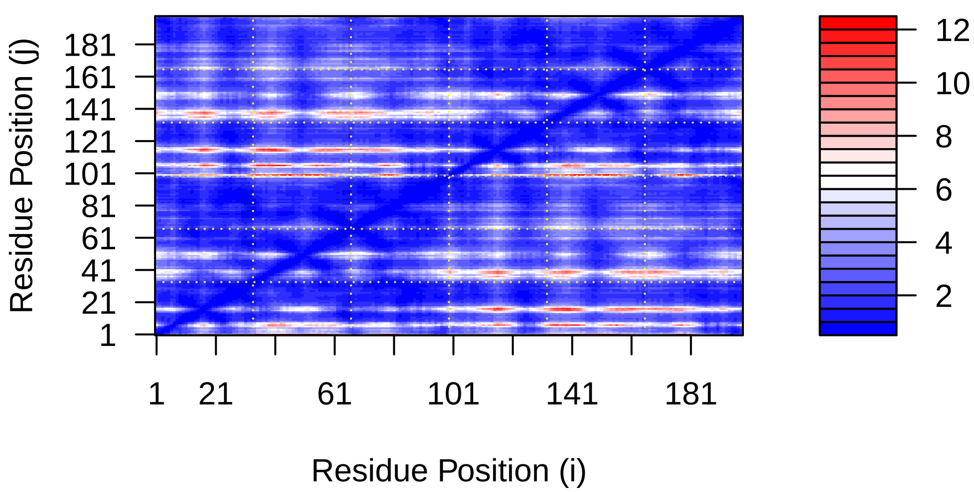
```
# Per-residue pLDDT scores
#  same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 91.62 94.06 94.56 93.88 96.12 90.69
```

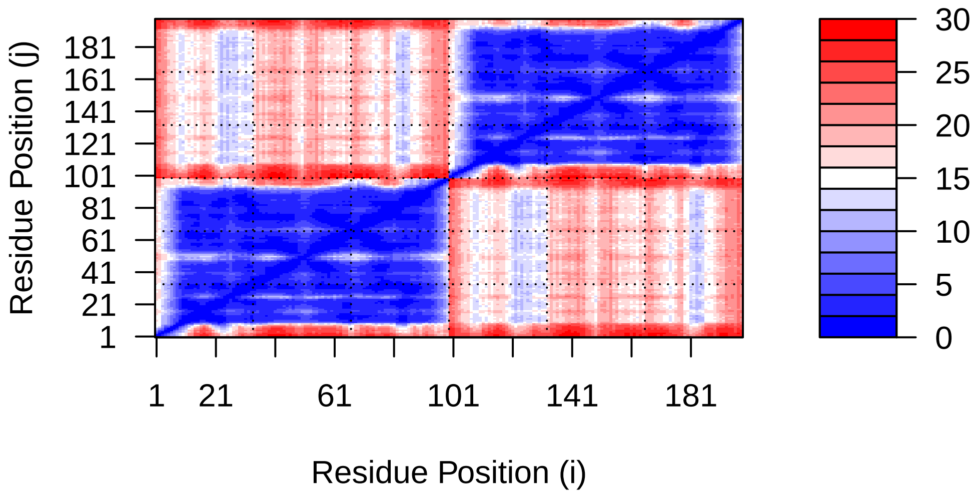
```
pae1$max_pae
```

```
[1] 12.33594
```

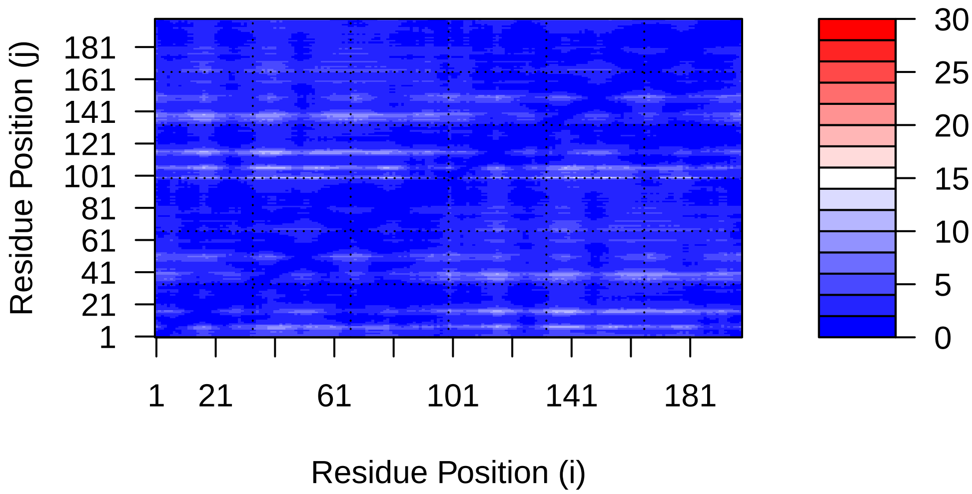
```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```




```
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```




```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Demonstrates error in alighnments, lower PAE = more confidence.

```
pae2 <- read_json(pae_files[2], simplifyVector = TRUE)
pae3 <- read_json(pae_files[3], simplifyVector = TRUE)
pae4 <- read_json(pae_files[4], simplifyVector = TRUE)

pae2$max_pae
```

```
[1] 18.67188
```

```
pae3$max_pae
```

```
[1] 14.39062
```

```
pae4$max_pae
```

```
[1] 29.57812
```

The PAE values show how model 5 is worse than model 1 as the PAE scores are higher for model 5. Lower scores are preffered. The other models their PAE scores are 18.67188 for model 2, 14.39062 for model 3, and 29.57812 for model 4.

```r
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "HIV_dimer_23119//HIV_dimer_23119.a3m"
```

```r
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

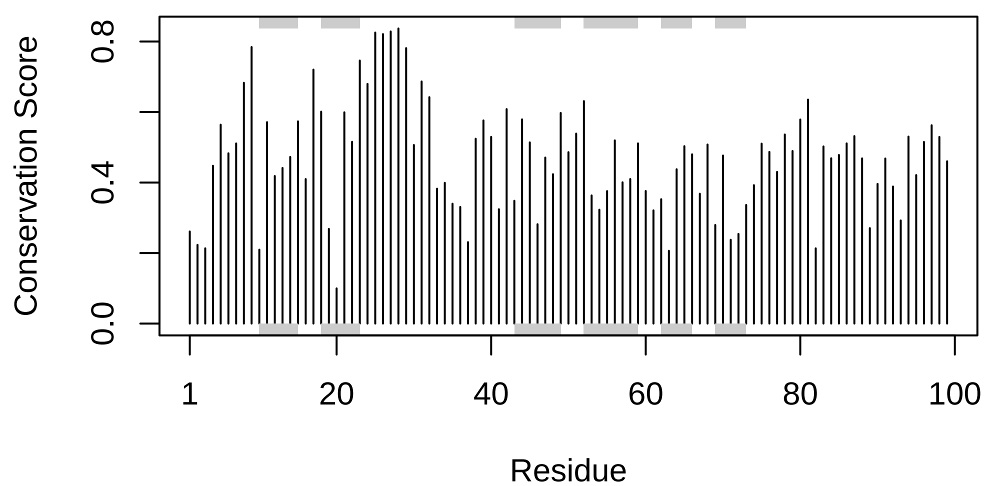
```r
dim(aln$ali)
```

```
[1] 5397  132
```

```r
sim <- conserv(aln)
```

Plot to demonstrate conservation, highly conserved residues are visualized through the graph.

```r
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```



```r
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```r
m1.pdb <- read.pdb(pdbfiles[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```